

1. Concept of Combining Multiple Perceptron for XOR-Gate Issue:

A. a single perceptron can't solve XOR-Gate issue, so combining two perceptrons and add another perceptron that uses the output of the two previous perceptrons as its inputs to create a two-level fully connected feedforward neural network for XOR-Gate issue.

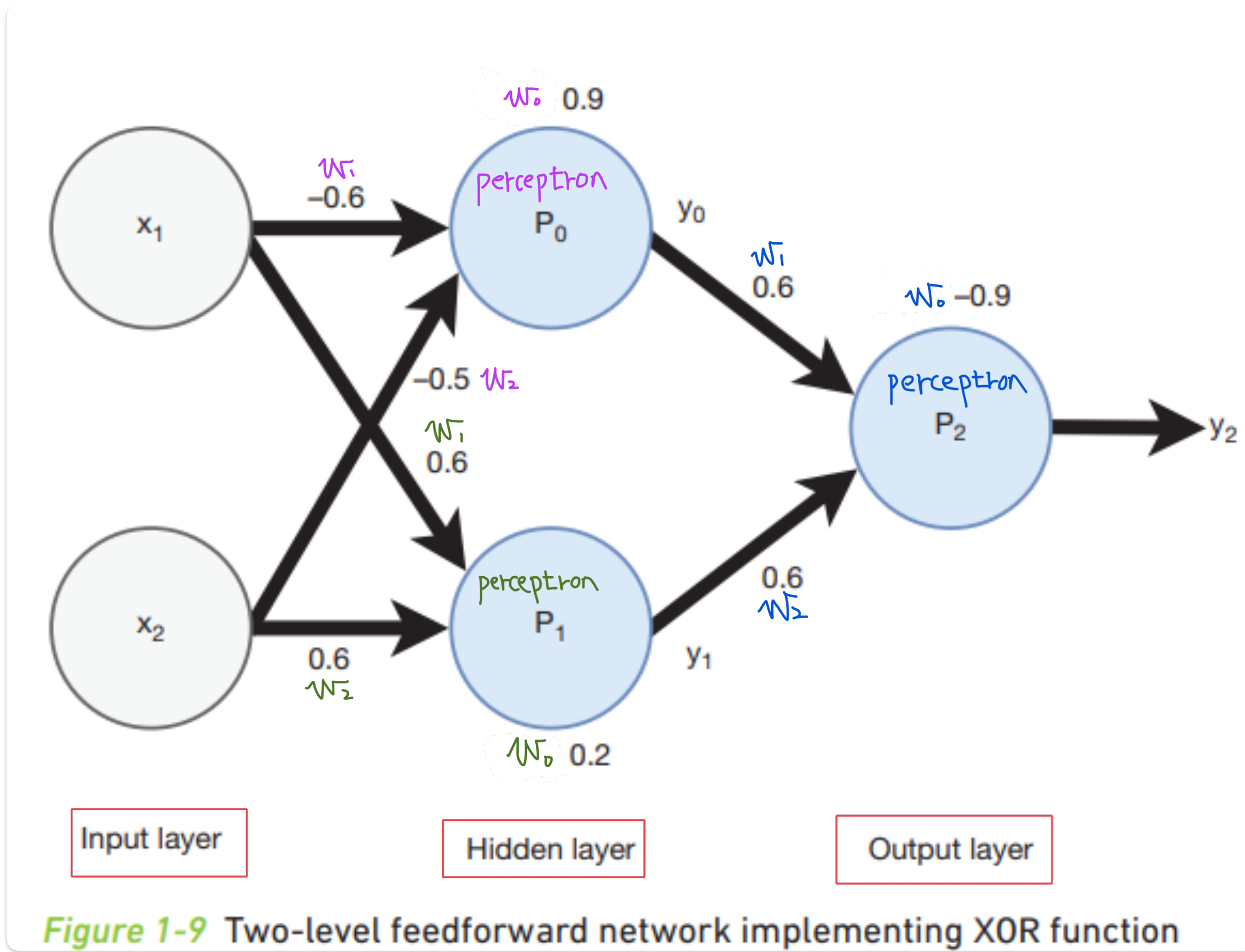
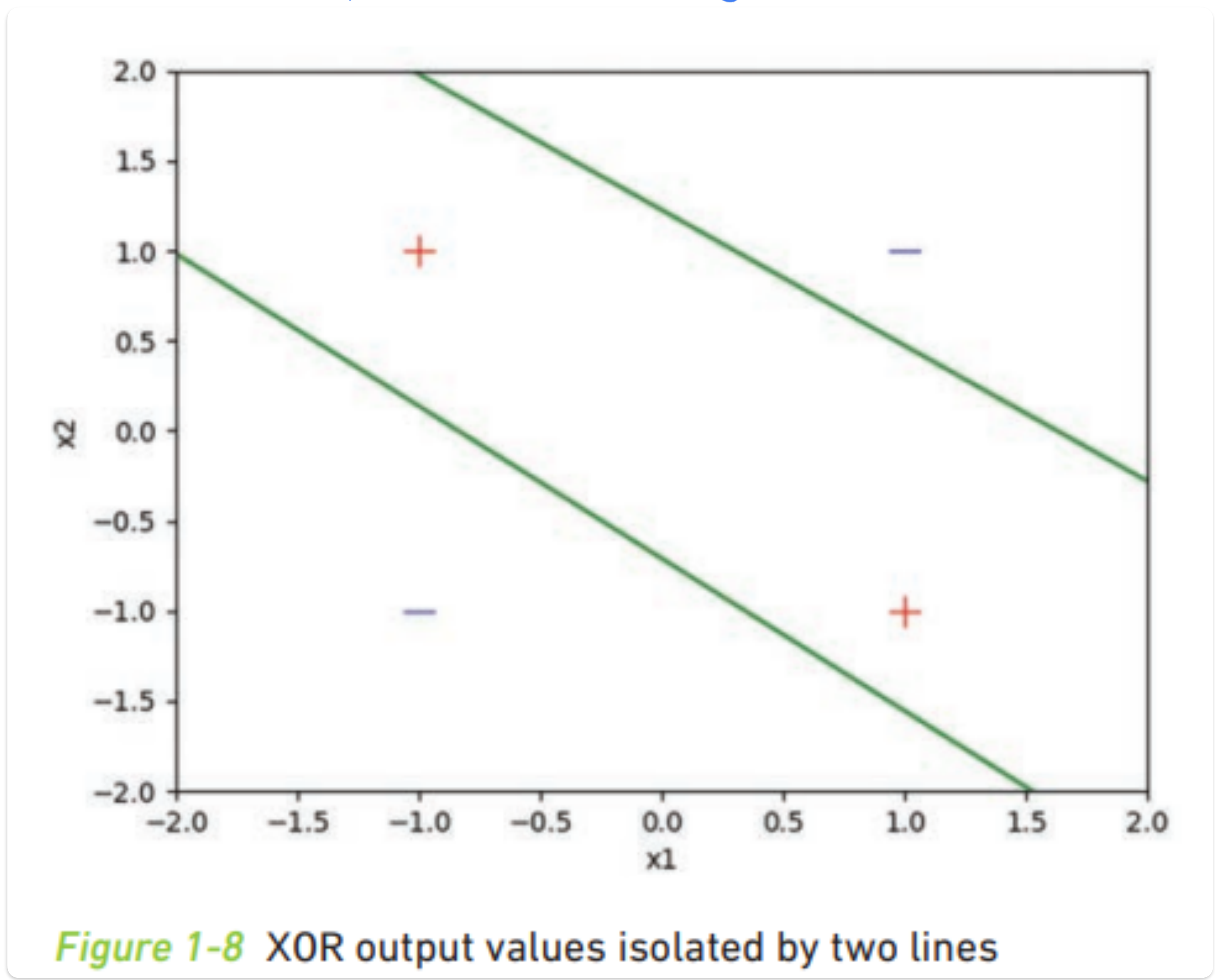


Table 1-3 Input and Output Values Showing That the Network Implements the XOR Function

X_0	X_1	X_2	Y_0	Y_1	Y_2
1	-1 (False)	-1 (False)	1.0 (True)	-1.0 (False)	-1.0 (False)
1	1 (True)	-1 (False)	1.0 (True)	1.0 (True)	1.0 (True)
1	-1 (False)	1 (True)	1.0 (True)	1.0 (True)	1.0 (True)
1	1 (True)	1 (True)	-1.0 (False)	1.0 (True)	-1.0 (False)

B. the architecture of two-level fully connected feedforward neural network for XOR-Gate issue can draw two straight lines in order to correctly separate all data points of pair training examples.



- C. perceptron P_0 in hidden layer implements a NAND Gate ($\overline{x_1 \cdot x_2}$)
perceptron P_1 in hidden layer implements a OR Gate ($x_1 + x_2$)
perceptron P_2 in output layer implements a AND Gate (NAND Gate) $\cdot$ (OR Gate)
combining outputs of P_1 and P_2 into input of P_2 , then perceptron P_2 in output layer implements a XOR Gate ($\overline{x_1 \cdot x_2} \cdot (x_1 + x_2)$)

2. Concept of Fully Connected Feedforward NN:

A. "fully connected" means the output of each perceptron in one layer is connected to the input of each perceptron in next layer.

B. "feedforward" means the neural network is a directed acyclic graph (DAG) and has no backward connection inside.
= a no cycle graph

3. Concept of Multilevel Perceptron NN:

A. fully connected feedforward NN is one kind of it.

B. it is a directed acyclic graph (DAG).
= a no cycle graph

- C. the structure
- input layer → inputting data into NN.
→ there is no perceptron in this layer.
 - hidden layer → placing between input layer and output layer.
→ outputs of perceptrons in hidden layer are not visible from outside of whole neural network.
→ DNN (Deep Neural Network) has multiple hidden layers in its structure.
 - output layer → outputting computing results from NN.
→ there are one or more than one perceptrons in this layer.